Urban Environmental Sound Classification Intelligence Report

Dataset File: UrbanSound8K.csv
Total Audio Samples: 8732
Number of Sound Classes: 10
Clustering Silhouette Score: 0.25
Execution Time: 8.67 seconds

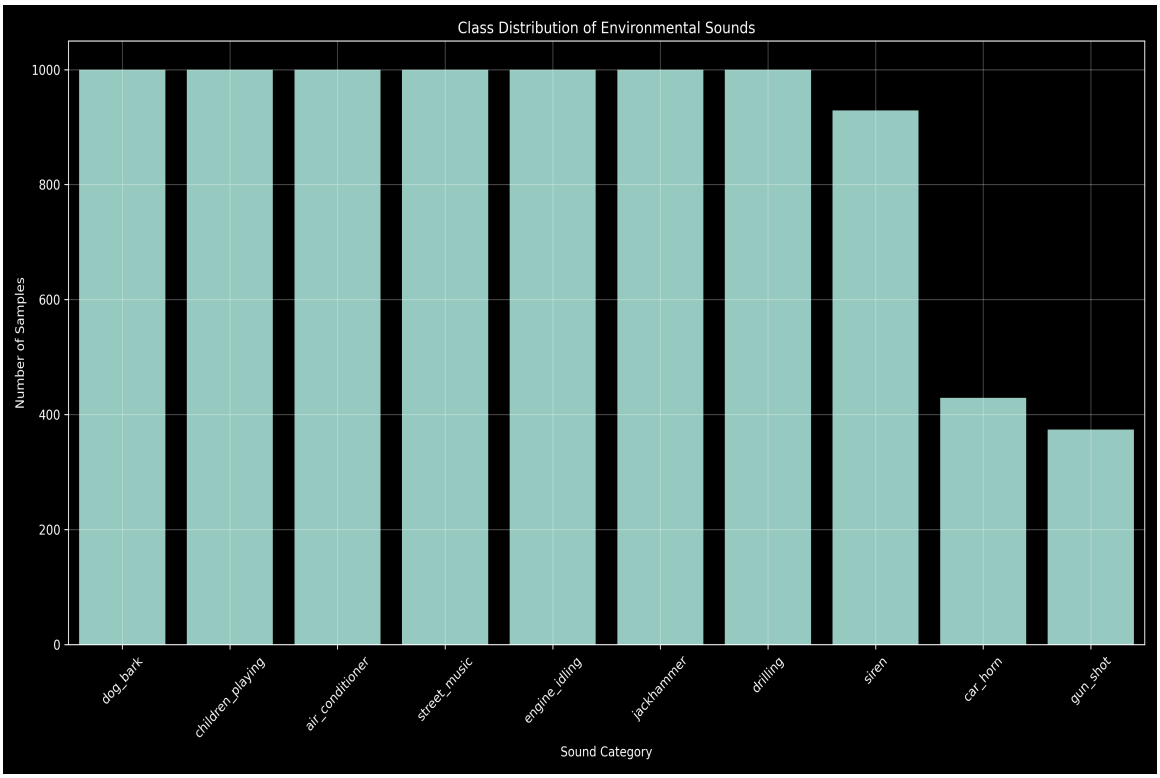
Model Performance Summary:

precision recall f1-score support

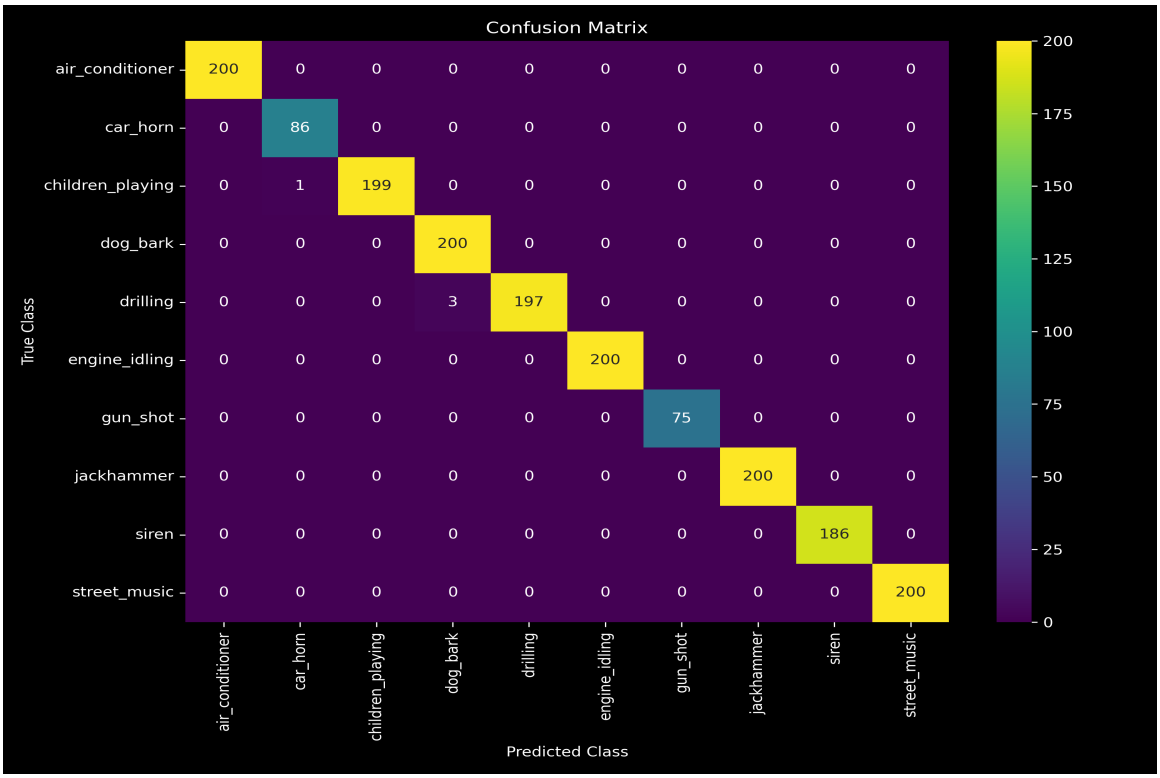
air_conditioner 1.00 1.00 1.00 200
car_horn 0.99 1.00 0.99 86
children_playing 1.00 0.99 1.00 200
dog_bark 0.99 1.00 0.99 200
drilling 1.00 0.98 0.99 200
engine_idling 1.00 1.00 1.00 200
gun_shot 1.00 1.00 1.00 75
jackhammer 1.00 1.00 1.00 200
siren 1.00 1.00 1.00 186
street_music 1.00 1.00 1.00 200

accuracy 1.00 1747
macro avg 1.00 1.00 1.00 1747
weighted avg 1.00 1.00 1.00 1747

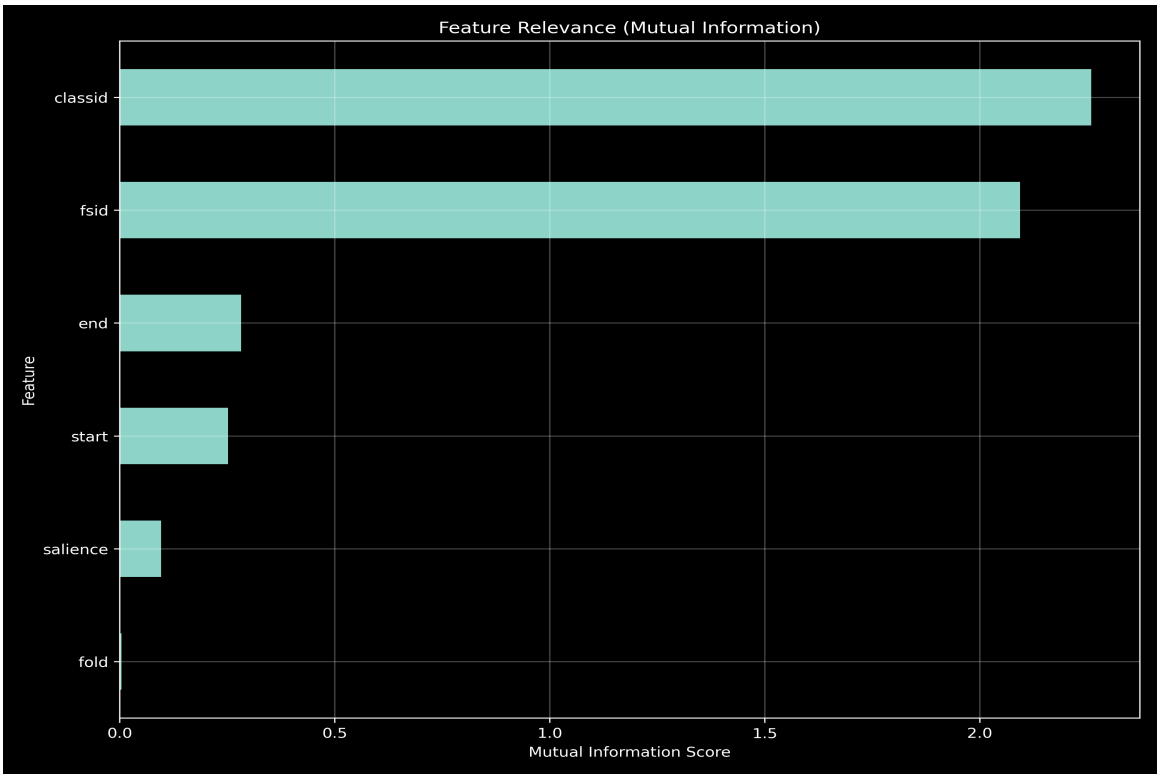
Class Distribution



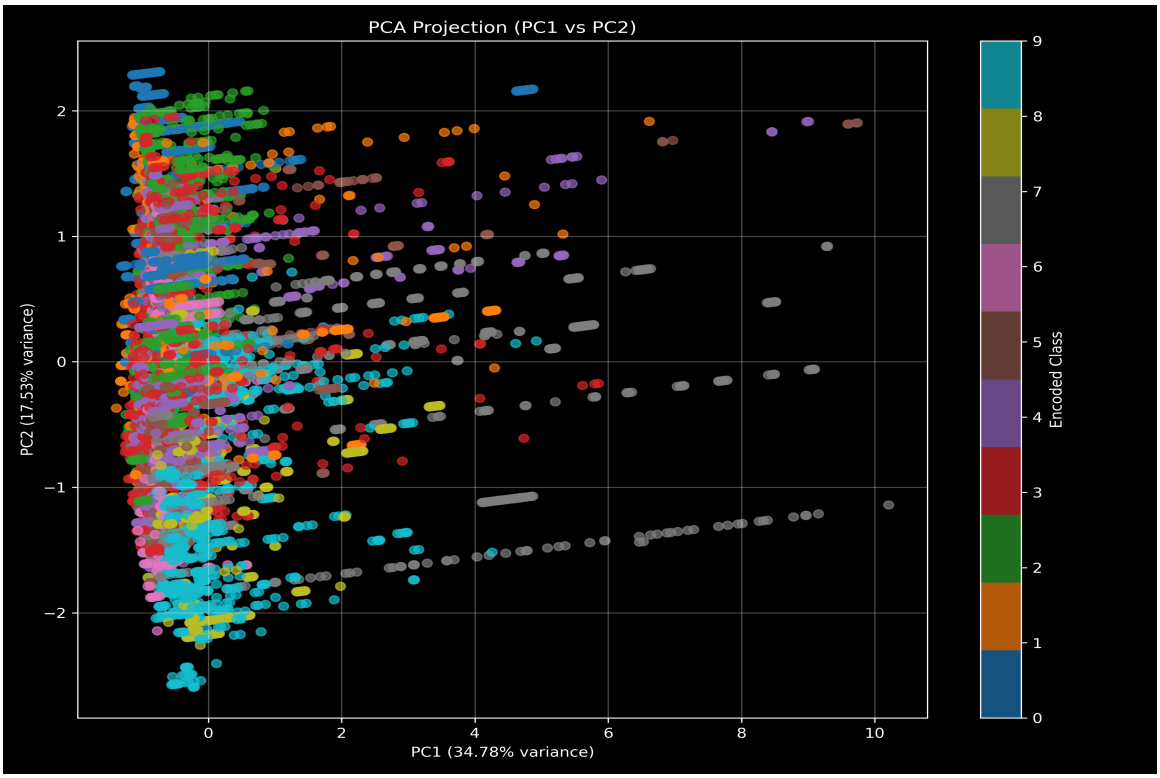
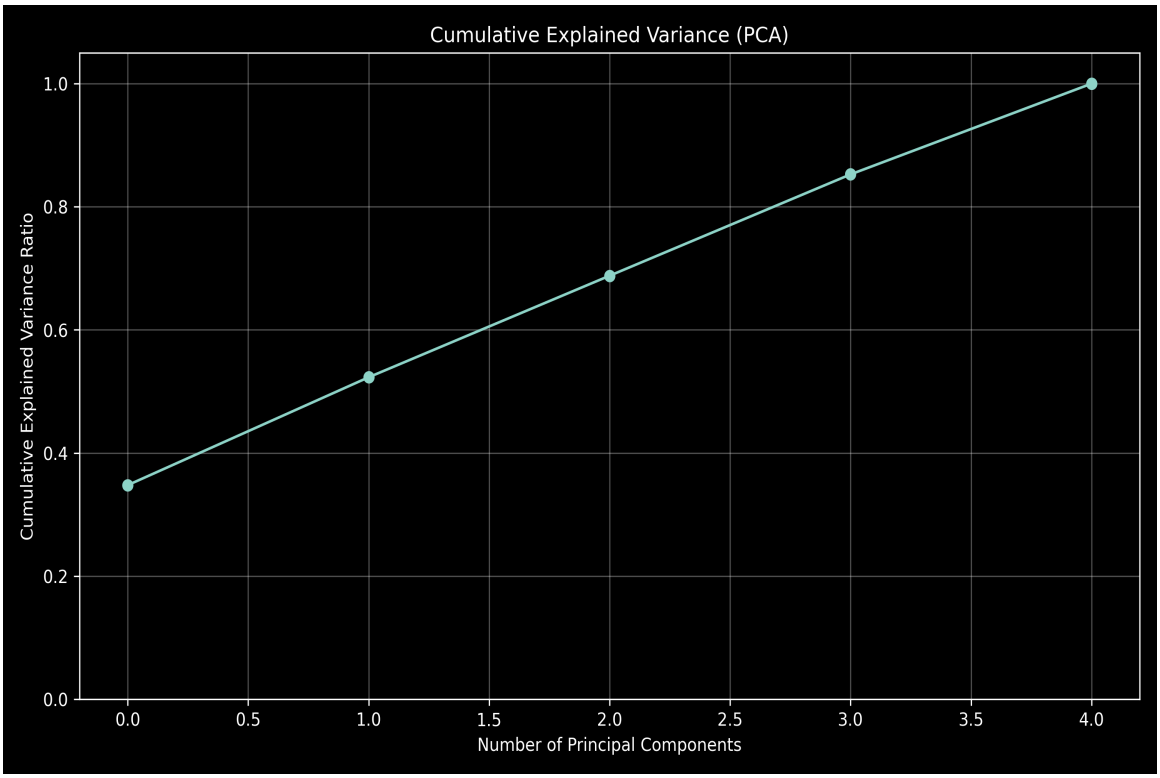
Confusion Matrix



Feature Relevance



Pca Projection



Pca Variance